

Kaan Berk Can

Game & Systems Designer · Technical Designer

Istanbul, Turkey · kaanc5528@gmail.com · +90 530 308 29 40
kaanberkcan-web-portfolio.vercel.app · linkedin.com/in/kaan-berk-can
Remote (UTC+3), flexible across EU, US and Asia hours · Open to relocation · Visa sponsorship required

PROFILE

Systems designer who balances games with the tools of an industrial engineer: mathematical models, simulation and data. Three years building crew-optimization software at Turkish Airlines, now completing an MSc in Game and Interaction Technologies at ITU. Portfolio includes a Monte Carlo-balanced card roguelike, an idle-economy study where human play beat Python optimizers, team-led playable demos in Unity and Godot, and complete GDDs for strategy and tabletop games.

SKILLS

Design	Systems & economy design · progression and balancing · technical design · level and puzzle design · GDDs and specs
Analysis	Mathematical modeling · Monte Carlo simulation · game analytics · Python · SQL · R · Excel
Engines & tools	Unity (C#) · Godot · Unreal Engine 5 · Git · Jira · Tabletop Simulator

GAME PROJECTS

Dekrawler [↗](#) Unity 6 · Solo · v0.4.0 on itch.io
Roguelike where a deck of cards is the dungeon. Weapons lock to the rank of what they kill; same-suit combos pay 1.5× per kill; a 13-level campaign built from rules, with 45 card-bound jokers and 15 rule-rewriting artifacts. Balanced with Monte Carlo runs that drive the real game build; interface drawn in code and every sound synthesized at runtime.

Idle Incrementation Analysis [↗](#) Unity · Excel · Python · Solo
900-second, six-tier idle economy ($\text{cost} = \text{base} \times 1.1^N$) modeled second by second. Human play beat greedy and brute-force Python optimizers, exposing local-optimum traps and combinatorial explosion; shipped as playable APK/EXE.

Harvey Park [↗](#) Godot · 4-person team, lead · Alpha demo
First-person psychological horror built on anomaly detection: two-phase room observation, binary elevator decisions across 16 floors, and a dual-threat pressure system (hidden timer plus a pursuer that freezes when watched).

Birth of Miracle [↗](#) Unity · 2–3 person team, lead · Vertical slice
2.5D cinematic puzzle platformer: environmental storytelling, timing and logic puzzles, perception-based abilities that reopen earlier areas, and a notebook system for narrative choices.

Miracle: Born & Dead [↗](#) Design document · Solo · 6+ months
Turn-based strategy with real-time tactical battles across five asymmetric factions; complete GDD.

Tabletop designs Board & megagames
Miracle: Board & Game (1–6 player co-op RPG on a 13×13 grid), Moods of Istanbul (4–16+ player civic megagame driven by Istanbul IPA data, run on Tabletop Simulator) and Mikapoly (co-op, no-elimination take on Monopoly).

Game concepts Pitch documents
Sown in Silence (farming sim by day, cosmic-horror exploration by night), Gears of Behemoth (boss-rush action scaling mechanical colossi) and Dissonance Ward (procedural 5×5 hospital horror with a multi-personality mechanic).

Player & design research Papers · video essays
Human vs generative-AI creativity in game design (first author); player perception across age, gender and play frequency; MDA analysis of Settlers of Catan; Bartle taxonomy case study of Path of Exile.

EXPERIENCE

Independent Game Developer & MSc Researcher 2024 – Present
Istanbul, Turkey
– **MSc thesis:** extracting procedural game mechanics from a given text and simulating them to test whether the text's emotional load can be conveyed through mechanics alone.
– Leads small teams as designer and developer on team projects (e.g. Harvey Park, 4-person team Boiler Works) and builds solo prototypes in Unity and Godot; regular game jam participant.

Turkish Airlines · Crew Planning / Systems Development Aug 2022 – Jul 2025
Istanbul, Turkey

Administrator / System Developer Oct 2024 – Jul 2025
– Developed and maintained the Jeppesen crew suite (Pairing, Rostering, Tracking, Manpower) while also running system and machine management with Bash, Oracle and MS SQL.
– Took part in the team-wide migration of the codebase from Mercurial to Git.

- System Developer

Apr 2023 – Oct 2024

 - **Crew assignment optimizer:** rewrote the assignment format and constraints from scratch, cutting run time by 10–25% while reducing rule violations and manual corrections by planners.
 - Built Rave rules, Python tooling and SQL / Power BI / Excel VBA reporting for planning of 20,000+ cockpit and cabin crew across the narrow- and wide-body fleet.
 - Worked directly with crew planners to turn operational requirements into rules and reports; tracked work in Jira.

- Intern Developer (part-time)

Aug 2022 – Apr 2023

 - Joined the same 20+ person team as a part-time developer while completing the internships required for the degree.

EDUCATION

MSc, Game and Interaction Technologies · Istanbul Technical University

2024 – exp. Dec 2026

BSc, Industrial Engineering (English) · Yıldız Technical University

GPA 3.46 / 4.00

2019 – 2023

ADDITIONAL

- Languages
- Turkish (native) · English (C1) · German (A1) · Japanese (A1)
- Training
- Jeppesen Rave training (Turkish Airlines)
- Community
- ITU Game Design & Development Club · game jam participant